24.2 PET Scanning

Question Paper

Course	CIE A Level Physics
Section	24. Medical Physics
Topic	24.2 PET Scanning
Difficulty	Easy

Time allowed: 40

Score: /33

Percentage: /100

Question 1a

State what is meant by the term *tracer* in relation to PET scanning.

[2 marks]

Question 1b

State the name of the particles that are emitted

(i)
by the tracer,

[1]

(ii)
from the body and detected by the detectors during PET scanning.

[1]

[2 marks]

Question 1c

Explain what is meant by *annihilation* in the context of PET scanning.

[2 marks]

Question 1d

State the **two** quantities which must be conserved in an annihilation event.

[2 marks]

Question 2a

Complete the following sentence about PET scanning

A positron is emitted by the radiotracer that interacts with an electron in the **detector / surrounding tissue** producing two γ-rays that move apart at **90° / 180°** to each other.

[2 marks]

Question 2b

The tracer commonly used in PET scanning is the isotope fluorine-18 $\left({}^{18}_{9}\mathrm{Fl}\right)$, which is a β^{+} emitter. The isotope is tagged onto glucose molecules before injecting it into the patient.

(i)
Write an equation for the decay of fluorine-18 $\left({}^{18}_{9}\mathrm{Fl}\right)$ into oxygen (symbol O).

[3]

(ii)
Glucose is a type of sugar found in many types of food. Suggest why the tracer is tagged onto glucose molecules.

[1]

[4 marks]

Question 2c

Fig. 1.1 shows a diagram of the components used in a PET scanner.

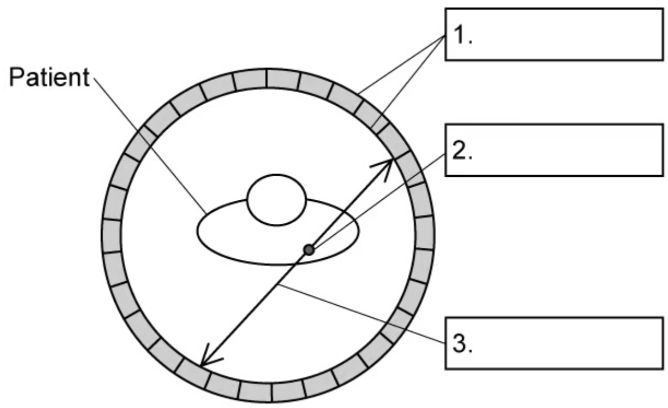


Fig. 1.1

Fill in the three missing labels on Fig. 1.1.

[3 marks]

Question 2d

Complete the following sentences to explain how gamma photons are used to produce an image of the tissue being studied. You may use a word more than once.

The gamma photons produced by the .... travel out of the body to opposite sides of the ....

A powerful computer uses the signals from the .... to trace the .... along which the .... occurred

The difference in .... between the two gamma rays is used to calculate the exact location the .... occurred

The relative number of gamma photons coming from each point enables the computer to determine the .... in each tissue in the patient

[4 marks]

Question 3a

Fill in the missing information in Table 1.1 about the properties of the particles involved in PET scanning.

Table 1.1

particle	mass / kg	charge / C
electron	9.11×10^{-31}	
positron		1.60×10^{-19}
gamma photon		

[3 marks]

Question 3b

When fluorine-18 decays, it emits a positron which only travels a very short distance before interacting with an electron, resulting in annihilation. As a result, gamma photons are produced.

Show that each gamma photon produced has an energy of 512 keV.

[4 marks]

Question 3c

Calculate the momentum, in N s, of one of the gamma photons produced in this annihilation.

[2 marks]

Question 3d

Fig. 1.1 shows a cross-sectional view of a patient's head inside a ring of gamma detectors during a PET scan.

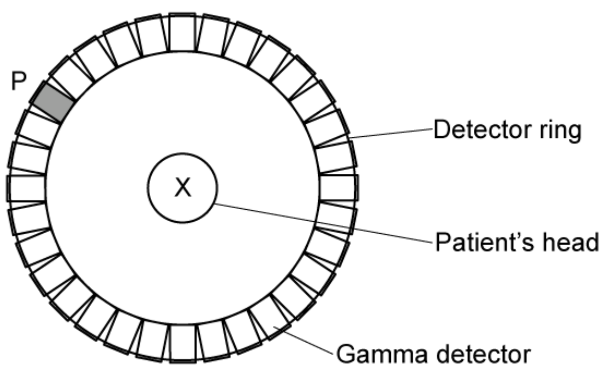


Fig. 1.1

A positron and an electron meet and annihilate at position X. Assume they have negligible kinetic energy when they meet.

Gamma photons are produced in this annihilation and are detected. The arrival of one gamma photon at detector P triggers a signal.

(i)

On Fig. 1.1, identify the other point on the detector that will be triggered by this annihilation.

[1]

(ii)

State and explain the conservation law which explains why P and the point you identified in **(b)(i)** will be triggered.

[2]

[3 marks]

Model Answers: Easy

1a

(a) A tracer is:

- A substance containing radioactive nuclei that are introduced into the body (before the PET scan); [1 mark]
- And are absorbed by the tissues (being studied); [1 mark]

[Total: 2 marks]

1b

(b)

(i) The particles that are emitted from the tracer are:

- Positrons; [1 mark]

(ii) The particles that are emitted from the body and detected by the detectors during PET scanning are:

- Gamma photons; [1 mark]

[Total: 2 marks]

1c

(c) Annihilation is:

- When a particle collides with its (corresponding) antiparticle so all its mass is converted into energy; [1 mark]
- (In PET scanning) a positron meets an electron and emits a pair of gamma photons; [1 mark]

[Total: 2 marks]

1d

(d) The quantities conserved in an annihilation event are:

- Mass-energy; [1 mark]
- Momentum; [1 mark]

[Total: 2 marks]

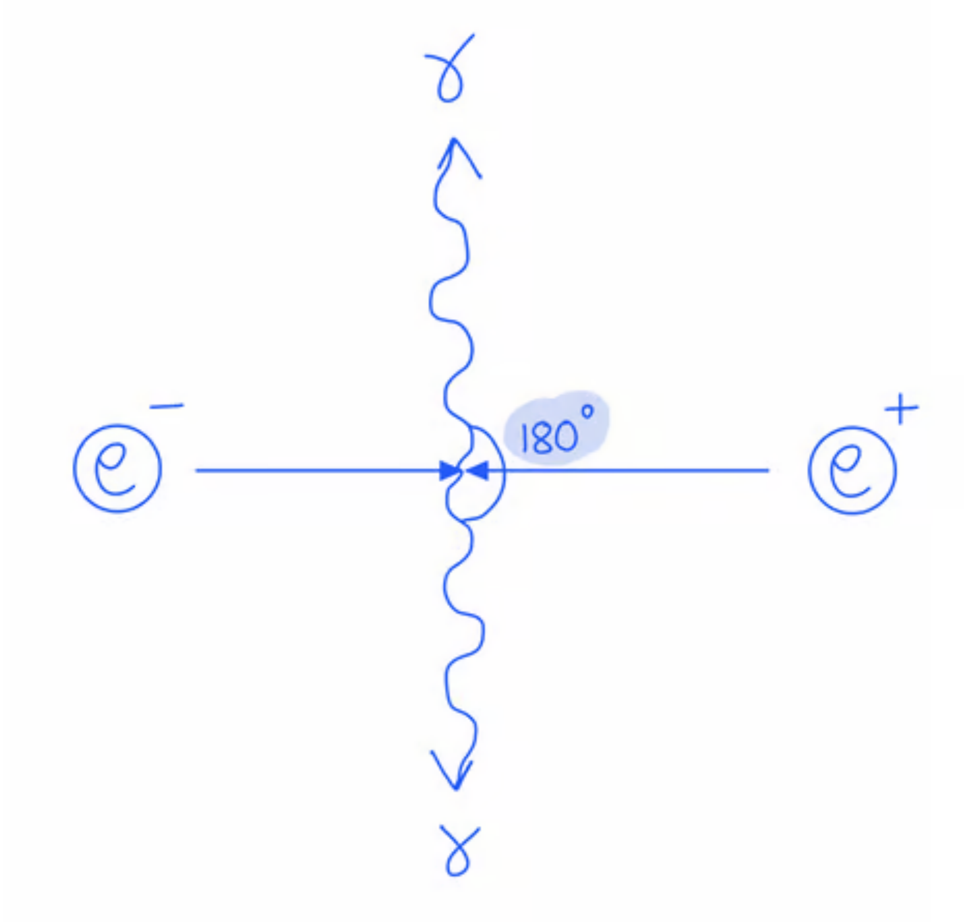
2a

(a) The correct sentence is:

- A positron is emitted by the radiotracer that interacts with an electron in the **surrounding tissue**; [1 mark]
- Producing two γ-rays that move apart at **180°** to each other; [1 mark]

[Total: 2 marks]

Momentum must be conserved in the interaction. Therefore, the gamma photons are emitted in opposite directions during annihilation.



2b

(b)

(i) The complete nuclear decay equation for fluorine-18 is:

- ${}^{18}_{9}\mathrm{Fl} \rightarrow {}^{18}_{8}\mathrm{O}$ [1 mark] + ${}^{0}_{+1}\beta$ [1 mark] + ν [1 mark]

(ii) The tracer is tagged onto glucose molecules because:

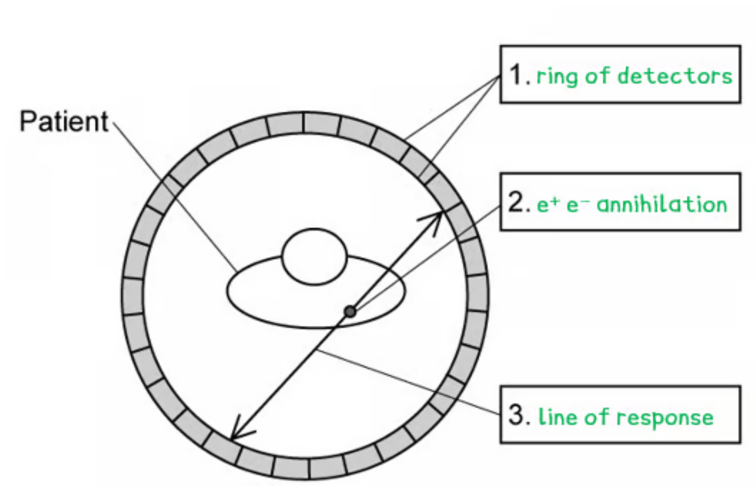
- Glucose is readily absorbed (by tissues); [1 mark]

[Total: 4 marks]

Remember that beta-minus particles are **electrons** whilst beta-plus particles are **positrons**. The tracer needs to emit positrons for them to annihilate with the electrons in the body.

2c

(c) The correct labels are as follows:



- *One mark for each correct label*

[Total: 3 marks]

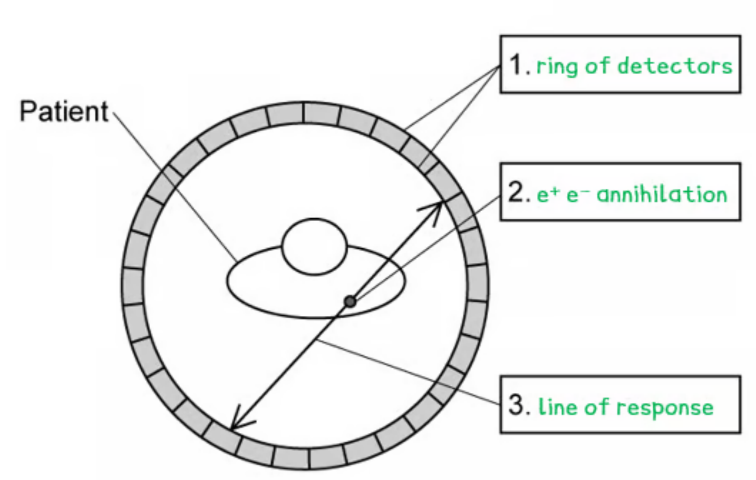
2d

(d) The correct sentences are:

- The gamma photons produced by the **electron–positron annihilation** travel out of the body to opposite sides of the **detector**; [1 mark]
- A powerful computer uses the signals from the **detector** to trace the **line of response** along which the **annihilation** occurred; [1 mark]
- The difference in **arrival time** between the two gamma rays is used to calculate the exact location the **annihilation** occurred; [1 mark]
- The relative number of gamma photons coming from each point enables the computer to determine the **tracer concentration** in each tissue in the patient; [1 mark]

[Total: 4 marks]

3a



- *One mark for each correct label*

[Total: 3 marks]

2d

(d) The correct sentences are:

- The gamma photons produced by the **electron-positron annihilation** travel out of the body to opposite sides of the **detector**; [1 mark]
- A powerful computer uses the signals from the **detector** to trace the **line of response** along which the **annihilation** occurred; [1 mark]
- The difference in **arrival time** between the two gamma rays is used to calculate the exact location the **annihilation** occurred; [1 mark]
- The relative number of gamma photons coming from each point enables the computer to determine the **tracer concentration** in each tissue in the patient; [1 mark]

[Total: 4 marks]

3a

(a) The correctly completed table is as follows:

particle	mass / kg	charge / C
electron	9.11×10^{-31}	**-1.60×10^{-19}**
positron	**9.11×10^{-31}**	$(+)1.60 \times 10^{-19}$
gamma photon	**0**	**0**

- *One mark for each correct row*

[Total: 3 marks]

3b

(b) Show that each gamma photon produced has an energy of 512 keV:

List the known quantities:

- Mass of an electron, $m = 9.11 \times 10^{-31}$ kg
- Speed of light, $c = 3.00 \times 10^{8}$ m s^{-1}
- 1 eV = 1.60×10^{-19} J

Convert the rest mass of an electron into rest energy:

- Mass-energy equivalence: $E = mc^2$ [1 mark]
- $E = (9.11 \times 10^{-31}) \times (3.00 \times 10^{8})^2$
- $E = 8.199 \times 10^{-14}$ J [1 mark]

Convert the energy in J to keV:

- $E = \frac{8.199 \times 10^{-14}}{1.60 \times 10^{-19}}$ [1 mark]

- E = 512 438 eV = 512 keV [1 mark]

[Total: 4 marks]

You could also convert 512 keV into SI units to show it is equal to the rest mass of an electron

$$E = 512 \text{ keV} \rightarrow (512 \times 10^{3}) \times (1.60 \times 10^{-19}) = 8.2 \times 10^{-14} \text{ J}$$

$$m = \frac{E}{c^2} = \frac{8.2 \times 10^{-14}}{(3.00 \times 10^{8})^2} = 9.11 \times 10^{-31} \text{ kg}$$

The value for the rest mass of an electron and the speed of light are given on your data sheet, so you don't need to memorise these!

3c

(c) Calculate the momentum of one of the gamma photons produced:

List the known quantities:

- Energy of each photon, $E = 8.2 \times 10^{-14}$ J (from **(b)**)
- Speed of light, $c = 3.00 \times 10^{8}$ m s^{-1}

Use the equation for photon momentum:

- Photon momentum: $p = \frac{E}{c}$ [1 mark]

- $p = \frac{8.2 \times 10^{-14}}{3.00 \times 10^{8}}$

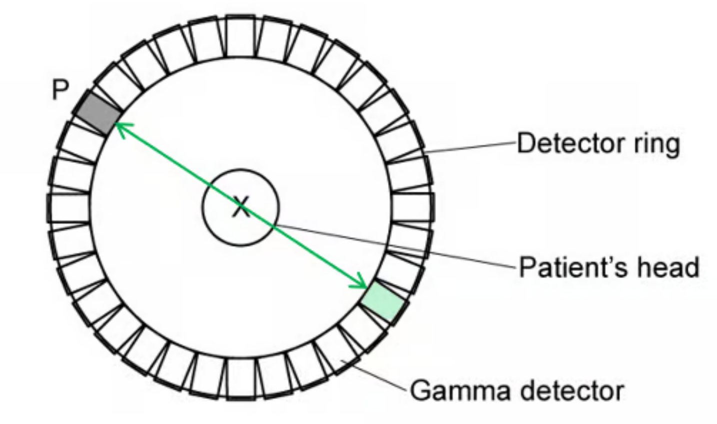
- $p = 2.7 \times 10^{-22}$ N s [1 mark]

[Total: 2 marks]

3d

(d)

(i) The other point on the detector that will be triggered is:



- Point exactly opposite P marked or shaded; [1 mark]

Use the equation for photon momentum:

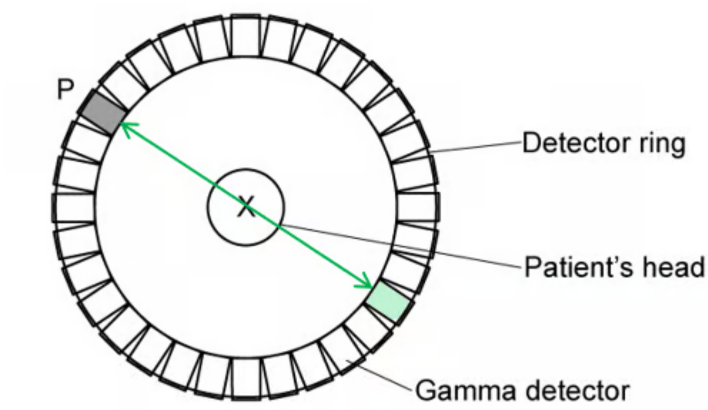
- Photon momentum: $p = \frac{E}{c}$ [1 mark]
- $p = \frac{8.2 \times 10^{-14}}{3.00 \times 10^{8}}$
- $p = 2.7 \times 10^{-22}$ N s [1 mark]

[Total: 2 marks]

3d

(d)

(i) The other point on the detector that will be triggered is:



- Point exactly opposite P marked or shaded; [1 mark]

(ii) Two gamma rays must be produced from electron-positron annihilation because:

- Momentum must be conserved; [1 mark]
- (Hence) two gamma rays are produced with equal and opposite momentum; [1 mark]

[Total: 3 marks]